

University Physics Quiz 4

Name: _____ SID: _____ School/Department: _____

1. The half life of a π^+ meson is 2.5×10^{-8} s. A beam of π^+ mesons is generated at a point 15 meters from a detector. Only $1/2$ of the π^+ mesons live to reach the detector. What is the speed of the π^+ mesons?

Solution1: Using the time dilation equation, the time a π^+ meson used to reach the detector in the detector frame is

$$t = \gamma t_0 = \frac{t_0}{\sqrt{1-\beta^2}}.$$

So the speed is $v = \frac{x}{t} = \frac{x\sqrt{1-\beta^2}}{t_0}$ or $\beta = \frac{x}{ct_0}\sqrt{1-\beta^2} = 2\sqrt{1-\beta^2}$. Thus

$$\beta^2 = 4 - 4\beta^2, \quad \beta = \frac{2\sqrt{5}}{5}.$$

Solution2: In the π^+ meson frame, the distance to the detector is

$$x' = x\sqrt{1-\beta^2}.$$

The time a π^+ meson used to reach the detector is $t_0 = \frac{x\sqrt{1-\beta^2}}{v}$.

Thus $\beta = \frac{x}{ct_0}\sqrt{1-\beta^2} = 2\sqrt{1-\beta^2}$, $\beta^2 = 4 - 4\beta^2$, $\beta = \frac{2\sqrt{5}}{5}$.

2. Two events occur in reference frame S $2\mu\text{s}$ apart and at the same point in space. In reference frame S' , the distance between the two events is 800m .

(a) What is the time interval between the two events in reference frame S' ?

(b) What is the velocity of S' frame relative to S frame?

Solution: (a) The spacetime interval is unchanged: $(c\Delta t)^2 - (\Delta x)^2 = (c\Delta t')^2 - (\Delta x')^2$.

Thus $(c\Delta t')^2 = (\Delta x')^2 + (c\Delta t)^2 - (\Delta x)^2$ or $c\Delta t' = \sqrt{(\Delta x')^2 + (c\Delta t)^2} = 1000\text{m}$, $\Delta t' = \frac{10}{3}\mu\text{s}$.

(b) Using the Lorentz transformation $\Delta t' = \gamma\left(\Delta t - \frac{u}{c^2}\Delta x\right) = \gamma\Delta t$, $\gamma = \frac{\Delta t'}{\Delta t} = \frac{5}{3}$.

$$\beta = \sqrt{1 - \frac{1}{\gamma^2}} = \sqrt{1 - \left(\frac{3}{5}\right)^2} = \frac{4}{5}, \quad u = \frac{4}{5}c.$$

3. (a) If the kinetic energy of a particle of rest mass m_0 is equal to its rest energy m_0c^2 , how much is its momentum? How much is its speed?

(b) A photon strikes an electron (of mass m_e) that is initially at rest, creating an electron-positron pair. The photon is destroyed and the positron and two electrons move off at equal speeds along the initial direction of the photon. What is the energy of the photon?

Solution: (a) The mass-energy of the particle is $E = E_k + m_0c^2 = 2m_0c^2$. Using the mass-energy momentum relation

$E^2 = p^2c^2 + m_0^2c^4$, we have

$$4m_0^2c^4 = p^2c^2 + m_0^2c^4, \quad p^2c^2 = 3m_0^2c^4, \quad p = \sqrt{3}m_0c.$$

Since $E = mc^2 = 2m_0c^2$, $m = \frac{m_0}{\sqrt{1-\beta^2}} = 2m_0$. $\sqrt{1-\beta^2} = \frac{1}{2}$, $\beta = \frac{\sqrt{3}}{2}$.

(b) Let the energy of the photon be E_p , then its momentum is $p = E_p/c$. From the conservation of the momentum,

the momentum for each electron/positron after collision is $p/3 = E_p/3c$. From the conservation of energy, we have

$$E_p + m_e c^2 = 3 \sqrt{\left(\frac{E_p}{3c}\right)^2 c^2 + m_e^2 c^4} = 3 \sqrt{\frac{E_p^2}{9} + m_e^2 c^4}$$

or

$$E_p^2 + m_e^2 c^4 + 2m_e c^2 E_p = E_p^2 + 9m_e^2 c^4, \quad E_p = 4m_e c^2.$$

4. A spaceship is built with two identical lights, one at the front and one at the rear. The ship normally travels with speed v_0 relative to Earth. When the ship approaches Earth, the light appears green and the wavelength of the light is $500nm$. When the ship departs Earth, the light appears red and the wavelength of the light is $600nm$.

(a) What is the speed v_0 of the spaceship? What is the wavelength of the light emitted from the headlight or taillight?

(b) There are two identical spaceships traveling along a straight line in the same direction leaving Earth. The one ahead travels with speed v_0 relative to Earth. The one behind trying to overtake the other with speed v relative to Earth finds that the taillight of the ship ahead appears green and the wavelength of the light is $500nm$. What is the speed of the overtaking spaceship?

Solution: (a) $\beta_0 = \frac{v_0}{c}$ $\lambda_o = \sqrt{\frac{1-\beta_0}{1+\beta_0}} \lambda_e$ blueshift, (1) (O: observed e: emitted)

$$\lambda'_o = \sqrt{\frac{1+\beta_0}{1-\beta_0}} \lambda_e \quad \text{redshift.} \quad (2)$$

(1)/(2): $\frac{\lambda_o}{\lambda'_o} = \frac{1-\beta_0}{1+\beta_0} \Rightarrow \beta_0 = \frac{\lambda'_o - \lambda_o}{\lambda'_o + \lambda_o} = \frac{600nm - 500nm}{600nm + 500nm} = \frac{1}{11}, \quad v_0 = \frac{c}{11}.$

(1)*(2) $\lambda_o \lambda'_o = \lambda_e^2 \quad \lambda_e = \sqrt{\lambda_o \lambda'_o} = 10\sqrt{30}nm.$

(b) (Solution 1) Denoting the speed of the overtaking ship relative to the ahead ship as v' , then

$$\beta' = \frac{v'}{c} \quad \lambda''_o = \sqrt{\frac{1-\beta'}{1+\beta'}} \lambda_e, \quad \Rightarrow \beta' = \frac{\lambda_e^2 - \lambda_o'^2}{\lambda_e^2 + \lambda_o'^2} = \frac{\lambda_o \lambda'_o - \lambda_o'^2}{\lambda_o \lambda'_o + \lambda_o'^2} = \frac{1}{11}. \quad v' = \frac{c}{11}.$$

Using the lorentz velocity transformation, the speed of the overtaking ship relative Earth is

$$v = \frac{v' + v_0}{1 + \frac{v' v_0}{c^2}} = \frac{\frac{2c}{11}}{1 + \frac{1}{11^2}} = \frac{11}{61}c.$$

(Solution2) From (a), we know that the wavelength of the light from the ahead ship for the observer in the reference frame of Earth is $600nm$. Then, using the Doppler blueshift equation, we have

$$500nm = \sqrt{\frac{1-\beta}{1+\beta}} 600nm, \quad \beta = \frac{v}{c}. \quad \Rightarrow \quad \beta = \frac{600^2 - 500^2}{600^2 + 500^2} = \frac{11}{61}, \quad v = \frac{11}{61}c.$$